

Python Project — Mini Supermarket Management System

Build a **Mini Supermarket Management System in Python** using the concepts you have learned so far: variables, lists, dictionaries, loops, conditions, and functions. The program should store a list of products with their prices, allow a customer to select products and quantities, calculate the shopping total, apply a 10% discount when the purchase is Rs. 1,000 or more, and generate a simple bill. You should also include basic sales analysis using sample sales data to identify total items sold and the best-selling product. Organize your program using separate functions for tasks such as displaying products, adding items to the cart, calculating the total, applying discounts, and generating the bill. The program should use a simple menu and handle invalid inputs properly.

Optional Advanced Features: If you complete the main Python project, add **CRUD functionality** to allow the supermarket manager to Create, Read, Update, and Delete products, including changing product prices and stock. As a further optional challenge, create a **Streamlit dashboard** for your supermarket system. The dashboard should display key information such as total products, total stock, total sales/revenue, best-selling products, and simple charts for sales and inventory. Advanced students can also add CRUD controls directly to the Streamlit dashboard so products can be added, updated, or deleted from the web interface.